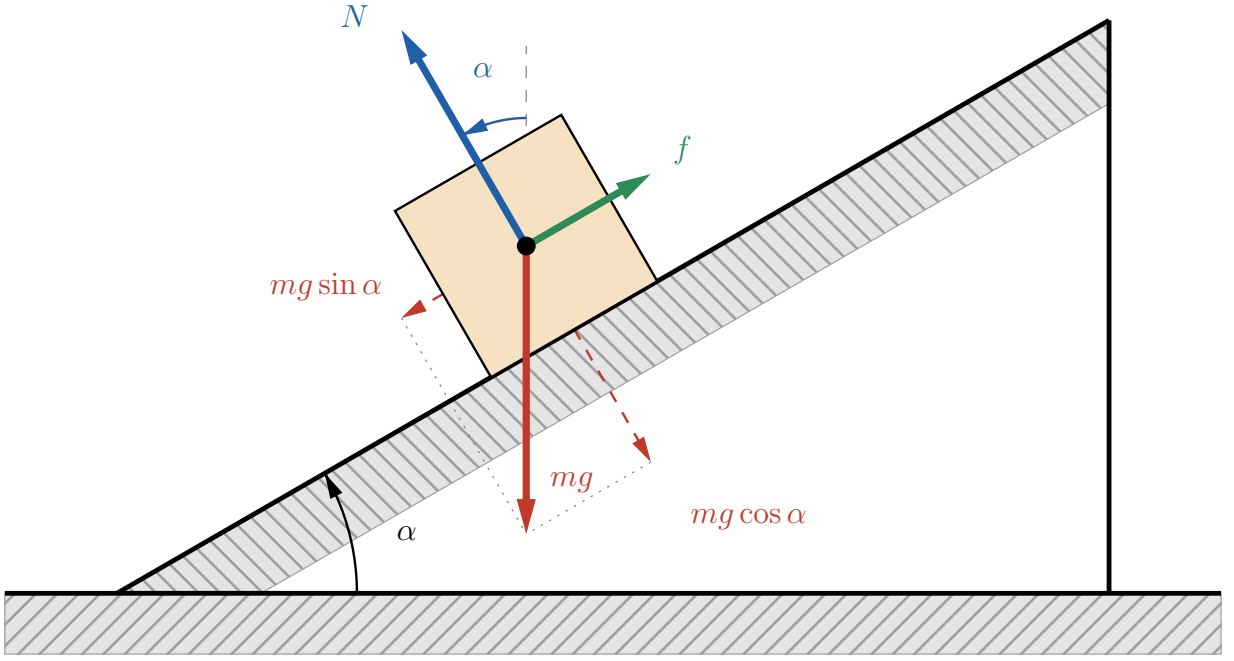
Block on an inclined plane

Free-body diagram of a block at rest on a 30° incline



At rest:

$N = mg\cos\alpha$

$f = mg\sin\alpha$

$f \leq \mu_s N$

$\Rightarrow \tan\alpha \leq \mu_s$

μ_s: coefficient of static friction